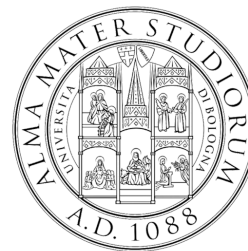


Construction and Validation of an Hidden Markov Model Profile for Kunitz Domain



Simona Ramella, Master in Bioinformatics - Unibo

Abstract

The Kunitz domain family comprehend protein structural motifs that act as serine protease inhibitors. Their function gives them an important role in biological processes such as hemostass, but are also used as toxins by snakes, scorpions and ticks.

In this report is described the workflow used for the construction of an Hidden Markov Model (HMM) profile for the Kunitz domain. The model has been trained over a Multiple Sequence Alignment (MSA) of a cohort of representative sequences retrieved from PDB and later filtered. Then, it has been then tested against a unified dataset of positive and negative sequences downloaded from Uniprot. The testing ables for the evaluation of the performance of the model through the Matthews Correlation Coefficient (MCC) and the selection of threshold which minimises the error.

Introduction

The Kunitz domain was firstly isolated from the Bovine Pancreatic Trypsin Inhibitor. Later on, they where found to be extremely represented across nature, with the main function of serine proteases inhibitor. A Kunitz domain is usually long between 50 and 60 amino acids and weights about 7 kDa. Its sequence folds into two anti-parallel β -sheets and one terminal α -helix. This structure is stabilised by three disulphide bridges created between six conserved Cysteines. C1-C6 connects N terminal and C terminal portion of the sequence, C2-C4 stabilises and properly orients the active inhibitor site at the protein surface, C3-C5 links the beta-sheet core to the alpha-helix region.

The domain works by mimicking a substrate of serine proteases. The active site of the domain is composed by a prominent reactive site loop (RLP) which acts as a rigid pseudo-substrate: it enters the active site of the protein and binds it tightly, preventing the enzyme from further catalytic activity. Usually the RLP spans from residue 11 to residue 19 and is held in place by the C2-C4 disulphide bond. The specificity of the domain is determined by the amino acid present at position 15, called position P1.

Given the structural importance of the disulphide bonds, the presence and relative position of the cysteines in the sequence are

expected to be highly conserved among Kunitz domain-containing proteins. Nevertheless, exceptions exist: in some cases, such as protein Conknitzin-S1 from cone snail venom, C2-C4 bond is missing and it is substituted by non-covalent alternative bonds carried by other amino acids like glycine.

A domain family is a cohort of different sequences with a common motif. This lets them share a similar function in different biological contexts. Therefore, for domain annotation, a simple pairwise alignment is not possible. Hidden Markov Models, on the other hand, are widely used, because of their efficiency in representing position-specific conservation patterns, insertions and delations.

The project aims to present a reproducible workflow for the construction and validation of a Hidden Markov Model for Kunitz protein domain.

Methods

Data Preparation for HMM Construction

The data for the construction of the HMM are retrieved from the protein data bank (PDB). It

is downloaded a costume cvs report of all protein containing the Kunitz Domain (Pfam: PF00014), with Polymer Entity Sequence Length between 30 and 70 and with Data Collation Resolution lower then 3,5 Armstrong. For each output the table reports: Protein ID (indicated as Entry ID), Sequence and Auth Asym ID (which indicated the chain of the protein).

The cvs is then processed and filtered (Supplementary materials: './scripts/consensus_screening.py'): sequences with length between 40 and 80 and containing the consensus pattern relative to the Kunitz Domain (F*Y*GC****N*F*****C) are kept. the total number of sequences considered is now 78.

To better represent the variability among the set and avoid redundancy 10 to 20 representative sequences will be extracted. This is performed using the MMseq2 tool where Minimum sequence identity and Minimum alignment coverage are set to 0.95. 13 representative sequences are selected. The representatives sequences are structurally aligned using the PBDeFold software for multiple sequence alignment. The HMMER software is used to build the model, taking as input the alignment fasta file. The Hidden Markov Model will then be validated.

Data Preparation for HMM Validation

For the evaluation of the performance of the model, two validation set are constructed from a positive and negative set.

In Uniprot is used the query (reviewed:true) AND (xref:pfam-PF000014) for collecting the sequences containing the Kunitz domain: this will form the positive dataset with 398 total sequences. On the other hand, with the query (reviewed:true) NOT (xref:pfam-PF000014) are retrieved the sequences that do not contain the Kunitz domain: it will form the negative dataset of 574229 sequences.

With the HMMER software, the model is used over both set of sequences.

For the positive set, all sequences are found to contain, with a certain probability, the Kunitz domain. To summarise the output is

created a file containing for each row: protein ID, E-value, and true label of 1. The positive set is shuffled. (Supplementary Materials: './output_val/positive_kunitz.match').

For the negative set, 5730 are found to contain the Kunitz domain with a certain probability. The same file type is created, but the true label is 0. The leaving 568499 proteins from the negative set are added to the file with an artificial e-value of 100 and a true label of 0. The negative set is shuffled.

(Supplementary Materials: './output_val/negative_kunitz.tot.match').

The first validation set contains the first 199 sequences of the positive set and first 287115 sequences of the negative one.

The second validation set contains the last 199 sequences of the positive set and last 287115 sequences of the negative one (Supplementary Materials: './val_sets/kunitz_set_1.txt' and './val_sets/kunitz_set_2.txt').

HMM Performance Evaluation

The performance of the model is evaluated on both sets for thresholds ranging from 1e-1 to 1e-15, using a python script (Supplementary Materials: './scripts/performance.py'). The script calculates the confusion matrix for each threshold and the respective evaluation parameters: accuracy, MCC, sensitivity, specificity, FPR, Precision and F1 score. (Supplementary Materials: './val_sets/kunitz_set_1.results' and './val_sets/kunitz_set_2.results').

Results

Logo Analysis

The skylign shown in figure 1 is constructed over the HMM model. The 6 Cysteines known to hold in place the structure of the domain, as well as the amino acid which compose the consensus motif of the sequence, F*Y*GC****N*F*****C, are expected to be highly conserved and, in fact, are easily retractable over the sequence.

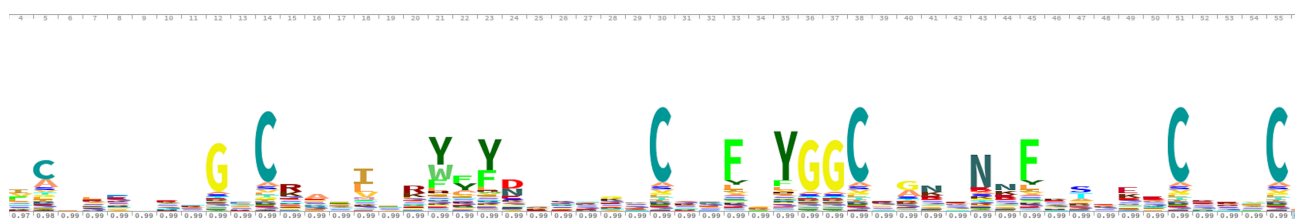


Figure 1: Skyline of Kunitz Domain constructed over its HMM

The pattern fully supports the validity of the selected sequences for the MSA in representing the Kunitz Domain.

ROC Curve

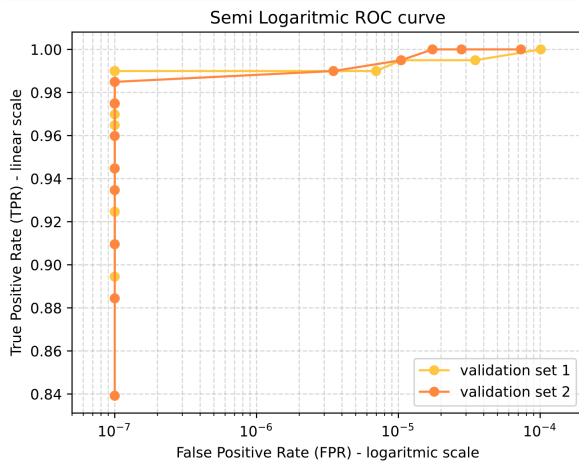


Figure 2: Semi logarithmic ROC curve validation set 1 (yellow line) and validation set 2 (orange line)

To understand the behaviour of the data at different thresholds, is generated a semi-logarithmic ROC curve. The plot clearly shows that the model is able to maintain an high true positive rate for all tested threshold. Furthermore, the results appear to be consistent in both validation set, indicating a good performance of the model across different set of data.

Selected Threshold

The performance of the model is evaluated for thresholds ranging from $1e-1$ to $1e-15$. The selected threshold should minimise the number of errors and consequentially maximise the MCC value.

For the first validation set the minimum number of errors is observed from $1e-5$ to $1e-8$: in all these are detected 0 false positive and 2 false negative, all with a MCC of 0.9949587192192321.

Validation set 2 help us select a singular threshold value. In this case a minimum number of errors is observed at threshold $1e-6$ and $1e-5$, but just the latter had the slightly higher MCC of 0.9924412533799732.

A thresholds of $1e-5$ is able to guarantee an optimal performance in both datasets and is therefore selected as optimal. To understand the behaviour of the data at different thresholds.

Conclusions

The constructed HMM is able to correctly identify the Kunitz domain at a selected threshold of $1e-5$ with 99.99% accuracy. The model is shown to be efficient in big validation sets characterised by high class imbalance. This demonstrates that the structural alignment used successfully captured the relative conserved features of the domain,

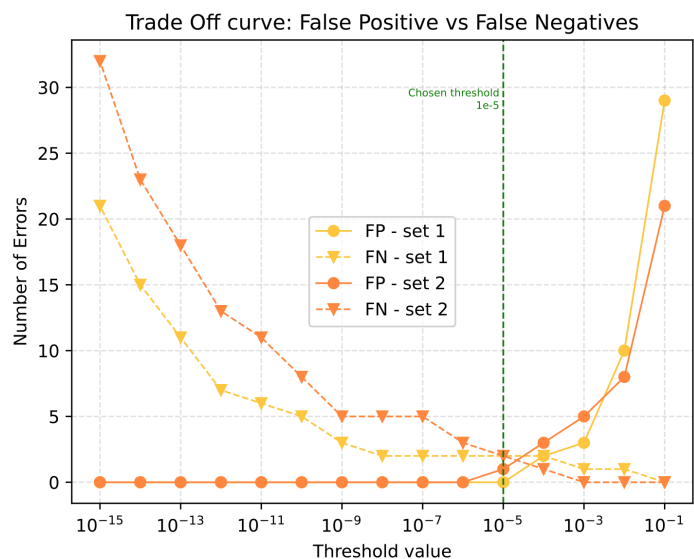
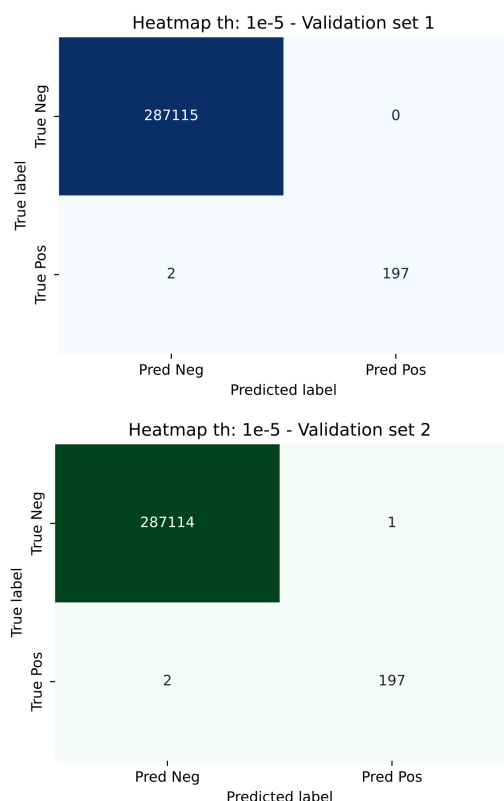


Figure 3- On the left: heatmaps of the confusion matrices of validation set 1 (blue) and validation set 2 (green), both for the selected threshold $1e-5$. On the top: Trade Off curve of false positive against false negative of validation set 1 (yellow) and 2 (orange). Green vertical line indicates the selected threshold value.

ability further confirmed by the associated logo (figure 1).

The Trade Off curve in figure 3 clearly shows the expected number of errors at each tested threshold. In both datasets, a more permissive critical value would lead to an higher number of false positives. On the contrary a smaller threshold would correlate with more false negatives. The false-negative sequences are likely to be genuine Kunitz-domain proteins that diverge substantially from the conserved pattern captured by the HMM profile, potentially because of greater evolutionary distance.

In conclusion the model demonstrates to have a good discrimination power between Kunitz and non-Kunitz domain.

References

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Supplementary Materials

All data, plots and script used for this project, can be found at this dedicated repository:

<https://github.com/simoram04-wq/HMM-for-kunitz-domain.git>